WIRSCHING'S CONJECTURE 3, AND THE PERIODIC CORRECTION TO BERG–KRÜPPEL'S DENSITY ASYMPTOTIC

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Abstract. Wirsching's 2003 argument for positive predecessor density under the $3n+1$ map reduces to three conjectures. Conjecture 3 concerns the near-zero asymptotics of an invariant density φ, compared against an explicit asymptotic φ_0 due to Berg and Krüppel. The exact log-Laplace transform of φ decomposes as a smooth part, a periodic correction H, and a doubly exponentially small remainder. Berg and Krüppel gave this correction, for this eigenfunction, as an infinite product in 1998, five years before Wirsching's conjecture; they did not ask whether it is constant. We give H instead as a Fourier series with closed-form coefficients in Γ and ζ: a classical zero-free theorem for ζ shows directly that it is not constant, and we rigorously enclose its oscillation. Wirsching's own comparison class fixes a single phase of H; there, Conjecture 3 holds, with an explicit limit. Away from that class the phase sweeps a full period, so the stronger, unrestricted asymptotic $\varphi(t) \sim \kappa\,\varphi_0(t)$ as $t \to 0^+$ fails. Wirsching needs less than the literal statement of Conjecture 3; we check directly that his actual requirement holds with a wide margin. This settles Conjecture 3 alone; Wirsching's other two conjectures, and positive predecessor density itself, remain open.

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1. Introduction

Let T denote the $3n+1$ map on the positive integers, $T(n) = n/2$ for n even and $T(n) = (3n+1)/2$ for n odd. Wirsching's monograph [3] develops the predecessor-set machinery this paper's argument descends from; [1] takes it further, studying the density of predecessor sets under T through an averaging construction on the 3-adic integers: a family of operators S_l, built from path counts in the predecessor graph, converges in the strong operator topology to a limiting operator S_∞ whose essential part is a Markov chain on $[0, 1]$. That chain has a unique invariant density φ, and Wirsching's proof of positive predecessor density reduces to three conjectures about φ and related quantities, stated as three unbridged gaps in the argument.

Of these three, this paper settles Conjecture 3. It concerns the behaviour of φ as its argument tends to 0, compared against an explicit comparison function φ_0: Wirsching's own equation (7.11) defines φ_0 to be exactly Berg and Krüppel's asymptotic solution, citing [2] by name at that point, so the identification is Wirsching's own choice, not a substitution made here. Berg and Krüppel had produced φ_0 five years earlier, by Laplace transform and saddlepoint analysis, as the asymptotic solution of a truncated version of the functional equation φ satisfies. Their argument for the asymptotic comparability of φ and φ_0 was informal: they give the periodic correction relating the two as an explicit infinite product (Corollary 3 below) but never evaluate it, so whether a single constant relates φ to φ_0 across every sequence, and whether the ratio converges at all along an arbitrary one, are both left open. Wirsching considers the strongest possible reading of that gap directly:

> "That replacement would be possible, if, e.g., $\varphi \sim c\,\varphi_0$ for some constant $c > 0$. It turns out that the following (slightly weaker) conjecture . . . suffices."

He then states ([1], immediately after the quoted sentence, with notation normalized: his own δ_5 is written δ throughout this paper, since no other δ-indexed class of his appears here, and his constant c, kept below for fidelity to his statement, is unrelated to $c := \log 3$ introduced in Section 3):

Wirsching's Conjecture 3. *There are positive real constants c and δ such that*

$$\lim_{l\to\infty} \frac{\varphi(z_l)}{\varphi_0(z_l)} = c > 0 \qquad \text{uniformly for sequences } (z_l) \in \widetilde{A}_\delta.$$

Here $\widetilde{A}_\delta$ is Wirsching's class of sequences tending to 0 at the borderline rate $z_l \sim l \cdot 3^{-l}$ (his equations (1.5) and (3.2), reproduced ahead of Theorem 1's proof): a sequence (x_l) belongs to $\widetilde{A}_\delta$ exactly when $\lfloor 3^l x_l \rfloor = k_l$ for a sequence of integers (k_l) with $|l - k_l| \le \delta\sqrt{l}$ for every l. This is weaker than the unrestricted asymptotic he first considers. The constant cancels out of the one inequality his own proof actually needs, so a class-dependent limit is enough. Neither statement had been settled before the following two theorems.

Theorem 1 (Conjecture 3)**.** *Wirsching's Conjecture 3 (stated just above) is true. For every $\delta > 0$, on his comparison class $\widetilde{A}_\delta$, the ratio $\varphi(z_l)/\varphi_0(z_l)$ converges uniformly to $e^{H(0)}$, a well-defined real number by Theorem 4 regardless of its decimal expansion, evaluated here as $e^{H(0)} = 0.534122\ldots$ (a floating-point evaluation; Proposition 8's proof certifies the nearby difference $H(0) - H(\log \frac{3}{2})$ rigorously, but not $H(0)$ itself in isolation), where H is the periodic function of Theorem 4 below and the normalization of φ_0 is Berg and Krüppel's own, equation (9.6) of [2]. Conjecture 3 itself only asserts the existence of some positive constant, independently of normalization; the specific value $e^{H(0)}$ is tied to this particular normalization of φ_0.*

Theorem 2 (Failure of the unrestricted asymptotic)**.** *The stronger asymptotic $\varphi(t) \sim \kappa\,\varphi_0(t)$ as $t \to 0^+$, for any single constant κ and with no restriction to a comparison class, is false. The ratio $\varphi(t)/\varphi_0(t)$ oscillates indefinitely as $t \to 0^+$, with amplitude bounded below by a positive constant: it is governed by the exactly c-periodic function H evaluated at the saddle location $w_0(\tau)$, $\tau = \log(1/t)$, whose phase modulo c sweeps every value infinitely often as $\tau \to \infty$ (Lemma 16). That the amplitude is positive follows from a classical theorem, with no computation; its explicit numeric size is a computer-assisted result, a rigorous interval-arithmetic certificate (Proposition 8).*

A single computation underlies both theorems. Write $t = e^{-\tau}$ and let $H(w)$ be the $\log 3$-periodic function appearing in the exact decomposition of φ's log-Laplace transform (Theorem 4); the saddlepoint bridge of Sections 4, 5, and 6 shows $\log\varphi(t) - \log\varphi_0(t) = H(w_0(\tau)) + o(1)$, where the saddle location $w_0(\tau)$ advances by c per period of τ, up to a vanishing correction: $w_0(\tau{+}c) - w_0(\tau) = c + O(1/\tau)$ (Lemma 16). Wirsching's comparison class constrains $w_0(\tau_l)$'s phase modulo $\log 3$ to converge to a single value as $l \to \infty$; the unrestricted range $\tau \to \infty$ with no class restriction instead sweeps that phase through every value, since Φ_0 is a bijection (again Lemma 16). Whether the ratio converges or oscillates depends on which sequences one compares along, and the two theorems above read the same object at two different scopes.

Corollary 3. *e^H is Berg and Krüppel's own periodic factor for the eigenfunction φ, from Proposition 9.3 of [2], which they give as an infinite product; H itself is its logarithm. Its Fourier series, Proposition 6 below, is new. This corollary has two parts of different depth: that e^H agrees with their product* as a function*, proved in Remark 5 below by pure algebra ($e^H = e^{-Q}e^{L}e^{-\Delta}$ recombining Theorem 4's own pieces), and that this function is the periodic factor multiplying φ's own asymptotic, which needs the full saddlepoint chain of Theorem 13 and Propositions 17–18, proved below.*

H's construction, non-constancy, and certified oscillation are in Section 3; the saddlepoint transfer to φ itself and the identification with Berg and Krüppel's asymptotic, in Sections 4–6.

2. The invariant density and its Laplace transform

Let $U_1, U_2, \dots$ be independent, identically distributed uniform random variables on $[0,1]$, and set

$$X = \sum_{j\ge 1} \frac{2}{3^j} U_j \in [0,1].$$

The self-similar identity $X = \frac{2}{3}U_1 + \frac{1}{3}X'$, with X' an independent copy of X, shows that the density φ of X satisfies

$$\varphi(x) = \frac{3}{2}\int_{3x-2}^{3x} \varphi,$$

equivalently $\varphi'(x) = \frac{9}{2}\varphi(3x)$ on $(0, \frac{2}{3})$: the displayed integral equation is exactly [1]'s equation (6.1), $(W_3 f)(x) := \frac{3}{2}\int_{3x-2}^{3x} f$, and φ is exactly the function his Corollary 7 constructs as the unique L^1_{loc} solution of $\operatorname{supp}\varphi \subset [0,1]$, $\int_0^1 \varphi = 1$, $W_3\varphi = \varphi$, his invariant density, up to the reflection $x \mapsto 1-x$ fixing the convention. It is also a translate of Rvachev's atomic function h_3 [4], and, at dilation parameter $a = 3$ and $\lambda = 2/3$, an instance of Berg and Krüppel's family of functional equations $\lambda\,\psi'(t) = a\big(\psi(at) - \psi(at-2)\big)$. The reflection is in fact vacuous, $\varphi(x) = \varphi(1-x)$ identically: X and $1 - X = \sum_{j\ge 1} \frac{2}{3^j}(1-U_j)$ are equidistributed, since $1 - U_j \sim U_j$ independently for every j and $\sum_{j\ge 1} 2\cdot 3^{-j} = 1$. It is supported on $[0,1]$, continuous (by Lemma 11 below), and constant, equal to $3/2$, on the middle third $[1/3, 2/3]$.

For $s > 0$ write

$$K(s) = \log \mathbb{E}\big[e^{-sX}\big] = \sum_{j\ge 1} g\left(\frac{2s}{3^j}\right), \qquad g(b) = \log\frac{1-e^{-b}}{b}.$$

The branch of g is the one analytic on $\{\operatorname{Re} w > 0\}$ and real on $(0,\infty)$; since $(1-e^{-w})/w$ is analytic and non-vanishing there (its zeros lie on $2\pi i\mathbb{Z} \setminus \{0\}$, and $w = 0$ is a removable singularity with value 1), and the half-plane is simply connected, g exists and is unique. Differentiating termwise, justified because $g(w) = -w/2 + O(w^2)$ as $w \to 0$ so the tail of the series is normally convergent together with every derivative,

$$t(s) := -K'(s), \qquad V(s) := s^2 K''(s) > 0,$$

the second an exact positivity, $K''(s)$ being the variance of X under its exponential tilt by e^{-sX}. Since $t'(s) = -K''(s) < 0$, t is strictly decreasing on $(0,\infty)$; at the endpoints, $t(0^+) = \mathbb{E}[X] = \sum_{j\ge 1} 2\cdot 3^{-j}\cdot \frac{1}{2} = \frac{1}{2}$, and $t(s) \to 0$ as $s \to \infty$ because the exponential tilt concentrates on $\operatorname{ess\,inf} X = 0$ (each factor e^{-sX} suppresses any mass away from 0 as $s \to \infty$, and $X \ge 0$ a.s.). So $t(s)$ decreases strictly from $1/2$ to 0, and studying $\varphi(t(s))$ as $s \to \infty$ is the same as studying $\varphi(t)$ as $t \to 0^+$.

Berg and Krüppel's comparison function is the asymptotic solution of the truncated equation $\lambda\,\psi'(t) = a\,\psi(at)$, a here their dilation parameter (this letter is reused in Section 3 for this paper's own, unrelated linear coefficient of Q; the two never appear in the same formula except in Section 6, where the collision is flagged again at the point it matters), by Laplace transform and saddlepoint analysis. In their normalization (their equation (9.6), with $\alpha = \frac{1}{2} - \frac{\log 2}{\log 3}$, $\beta = \frac{1}{2\log 3}$, and γ, δ_{BK}, ε determined by α and β through their own equations, given explicitly in Section 6),

$$\varphi_0(t) = \frac{(2\beta)^{\varepsilon}}{\sqrt{2\pi}}\, t^{\gamma}\,(-\log t)^{\delta_{\mathrm{BK}}}\, \exp\left(-\beta\log^2\frac{t}{-\log t}\right).$$

Write $\varphi_{0,\mathrm{bare}}$ for the same expression without the leading constant $(2\beta)^{\varepsilon}/\sqrt{2\pi}$: the two differ only by that constant.

3. The periodic correction

Set $c = \log 3$ and write $w = \log s$, $L(w) = K(e^w)$. Substituting into the defining series and splitting off the $j = 1$ term against the rest reproduces the recurrence

$$L(w+c) - L(w) = \log\bigl(1 - e^{-2e^w}\bigr) - w - \log 2. \tag{1}$$

Let

$$Q(w) = -\frac{w^2}{2c} + \left(\frac{1}{2} - \frac{\log 2}{c}\right) w,$$

chosen so that $Q(w+c) - Q(w) = -w - \log 2$, matching (1)'s non-periodic terms exactly. Write $a := \frac{1}{2} - \frac{\log 2}{c}$ for Q's linear coefficient, used again from Section 5 onward.

Theorem 4. *Define*

$$H(w) := L(w) - Q(w) + \sum_{k\ge 0} \log\bigl(1 - e^{-2\cdot 3^k e^w}\bigr). \tag{2}$$

Then H is exactly c-periodic, $H(w+c) = H(w)$ for every $w \in \mathbb{R}$, and

$$L(w) = Q(w) + H(w) + \Delta(w), \qquad \Delta(w) = -\sum_{k\ge 0} \log\bigl(1 - e^{-2\cdot 3^k e^w}\bigr),$$

with $\Delta^{(j)}(w) = O_j\bigl(e^{jw - 2e^w}\bigr)$ for every $j \ge 0$: doubly exponentially small, together with all its derivatives, as $w \to +\infty$.

Proof. Write $d(w) = \log(1 - e^{-2e^w})$, so (1) reads $L(w+c) - L(w) = d(w) - w - \log 2$. Directly from the definition of Q (expand $(w+c)^2 = w^2 + 2wc + c^2$ and collect terms), $Q(w+c) - Q(w) = -w - \log 2$ as well, so the two non-periodic parts cancel exactly:

$$(L-Q)(w+c) - (L-Q)(w) = \bigl[d(w) - w - \log 2\bigr] - \bigl[-w - \log 2\bigr] = d(w).$$

The series in (2) telescopes this exactly: with $D(w) := (L-Q)(w)$,

$$D(w) + \sum_{k\ge 0} d(w+kc) = D(w+c) - d(w) + \sum_{k\ge 0} d(w+kc) = D(w+c) + \sum_{k\ge 0} d(w+(k+1)c),$$

so the right side of (2) is invariant under $w \mapsto w + c$, that is, H is c-periodic. Convergence of the series and its termwise differentiability, together with the stated bound on $\Delta^{(j)}$, follow from $d(w) = O(e^{-2e^w})$ and the corresponding bounds on its derivatives, since $2 \cdot 3^k e^w$ grows like 3^k for $k \ge 0$: the terms $d(w+kc)$ decay doubly exponentially in k, hence faster than any fixed geometric rate, which gives uniform, dominated convergence of the series and its derivatives for w in any half-line bounded away from $-\infty$. □

Given Q, H is uniquely determined by the requirement that $L - Q - H$ be $o(1)$ as $w \to \infty$: any two periodic functions differing from $L - Q$ by an $o(1)$ term as $w \to \infty$ must be equal, being periodic functions that agree in the limit along every residue class of w modulo c.

Remark 5. Berg and Krüppel's own Proposition 9.3, specialized to their dilation parameter 3 and their $b = 3/2$ (exactly this eigenfunction; their α, kept as a symbol here, is exactly Section 2's imported α, which Proposition 18 below identifies with this paper's own a), gives the equivalent identity $\exp(H(w)) = 3^{\frac{1}{2}t^2 - \alpha t} \prod_{k\ge 0} \frac{1 - e^{-3^{t-k}/b}}{3^{t-k}/b} \prod_{l\ge 1}\bigl(1 - e^{-3^{t+l}/b}\bigr)$, $t = w/c$, as an infinite product, in their proof of that proposition. This is $e^{-Q} e^{L} e^{-\Delta}$ term for term: writing $s = e^w = 3^t$, $3^{t-k}/b = 2s/3^{k+1}$, so reindexing $j := k + 1 \ge 1$ turns the first product into $\prod_{j\ge 1} \frac{1 - e^{-2s/3^j}}{2s/3^j}$, exactly $e^{L(w)}$

from K's defining series; $3^{t+l}/b = 2s\cdot 3^{l-1}$, so reindexing $k := l-1 \geq 0$ turns the second into $\prod_{k\geq 0}\big(1-e^{-2s\cdot 3^k}\big)$, exactly $e^{-\Delta(w)}$ by (2); and, since $w = ct$ and $e^c = 3$,

$$-Q(w) = \frac{w^2}{2c} - aw = c\big(\tfrac12 t^2 - at\big), \qquad \text{so} \qquad 3^{\frac12 t^2 - \alpha t} = e^{-Q(w)}$$

once $\alpha = a$ (Proposition 18 identifies Berg and Krüppel's α with this paper's own linear coefficient of Q, also called a). So Proposition 8's primary computation from (7) below is, in substance, an evaluation of this product. What Proposition 6's Fourier series supplies instead, and their product form does not, are the derivative bounds (5), on which the rest of the paper's saddlepoint estimates depend.

Proposition 6. *H has the Fourier expansion $H(w) = \sum_{m\in\mathbb{Z}} \hat H(m)e^{i\omega_m w}$, $\omega_m = 2\pi m/c$, with*

$$\hat H(0) = \frac{\log 2}{2} - \frac{\log^2 2}{2c} - \frac{c}{12} - \frac{A}{c}, \qquad A = \frac{\pi^2}{12} - \frac{\gamma_E^2}{2} - \gamma_1,$$

$$\hat H(m) = -\frac{2^{i\omega_m}}{c}\,\Gamma(-i\omega_m)\,\zeta(1-i\omega_m), \qquad m\neq 0,$$

where γ_E is the Euler–Mascheroni constant and γ_1 the first Stieltjes constant.

Proof. Both g and K have Mellin transforms on the strip $-1 < \operatorname{Re} z < 0$. As $b\to 0^+$, $g(b) = -b/2 + O(b^2)$; as $b\to\infty$, $g(b) = -\log b + O(e^{-b})$. So $\int_0^\infty |g(b)|\,b^{\sigma-1}\,db < \infty$ for $-1<\sigma<0$, and the same two bounds applied termwise give $K(s) = O(s)$ as $s\to 0^+$ and $K(s) = O(\log^2 s)$ as $s\to\infty$: the terms with $3^j \leq 2s$ number $O(\log s)$ and are each $O(\log s)$, and the terms with $3^j > 2s$ sum to $O(1)$.

Integrate by parts. With $g'(b) = 1/(e^b-1) - 1/b$, and both boundary terms vanishing on the strip ($g(b)b^z \to 0$ at 0 because $\operatorname{Re} z > -1$, and at ∞ because $\operatorname{Re} z < 0$),

$$g^*(z) = \int_0^\infty g(b)\,b^{z-1}\,db = -\frac1z\int_0^\infty\Big(\frac{1}{e^b-1}-\frac1b\Big)b^z\,db = -\frac{\Gamma(z+1)\zeta(z+1)}{z} = -\Gamma(z)\,\zeta(1+z),$$

the middle step being the classical continuation $\int_0^\infty\big(\frac{1}{e^u-1}-\frac1u\big)u^{v-1}\,du = \Gamma(v)\zeta(v)$, valid for $0 < \operatorname{Re} v < 1$, at $v = z+1$.

Now the harmonic sum. In the j-th term substitute $u = 2s/3^j$, so $s = 3^j u/2$ and $s^{z-1}\,ds = (3^j/2)^z u^{z-1}\,du$:

$$\int_0^\infty g\Big(\frac{2s}{3^j}\Big)s^{z-1}\,ds = \Big(\frac{3^j}{2}\Big)^z\int_0^\infty g(u)\,u^{z-1}\,du = \Big(\frac{3^j}{2}\Big)^z g^*(z).$$

Summing over $j\geq 1$ is legitimate termwise, since $\sum_{j\geq1}(3^j/2)^\sigma\int_0^\infty |g(u)|u^{\sigma-1}\,du$ converges for $-1<\sigma<0$. The multiplier that appears is therefore

$$\sum_{j\geq1}\Big(\frac{3^j}{2}\Big)^z = 2^{-z}\sum_{j\geq1}3^{jz} = \frac{2^{-z}3^z}{1-3^z} = \frac{(3/2)^z}{1-3^z},$$

a geometric series of ratio 3^z, of modulus $3^{\operatorname{Re} z} < 1$, convergent on $\operatorname{Re} z < 0$ and nowhere else. Hence

$$K^*(z) = \frac{(3/2)^z}{1-3^z}\cdot\big(-\Gamma(z)\zeta(1+z)\big) = \frac{(3/2)^z\,\Gamma(z)\,\zeta(1+z)}{3^z-1}, \qquad -1<\operatorname{Re} z<0. \tag{3}$$

The dilations $\mu_j = 2/3^j$ enter inverted, through μ_j^{-z}, which is where the 2^{-z} comes from, and with it the $2^{i\omega_m}$ of the statement.

In $\operatorname{Re} z > -1$ the right side of (3) is meromorphic, with a pole of order three at $z=0$ (each of $\Gamma(z)$, $\zeta(1+z)$ and $(3^z-1)^{-1}$ contributing one) and a simple pole at each remaining zero $z = i\omega_m$, $m\neq 0$, of 3^z-1. To the right of the imaginary axis it is analytic.

Fix $\sigma \in (-1,0)$ and $\sigma' \in (0,1)$, and shift the inversion contour in $K(s) = (2\pi i)^{-1}\int_{(\sigma)} K^*(z)s^{-z}\,dz$ from σ to σ', using rectangles with horizontal sides at $\operatorname{Im} z = \pm T_M$, $T_M := \pi(2M+1)/c$. On those sides $3^z = 3^{\operatorname{Re} z}e^{i\pi(2M+1)} = -3^{\operatorname{Re} z}$, so $|3^z - 1| = 1 + 3^{\operatorname{Re} z} \geq 1$; Stirling makes $|\Gamma(z)|$ decay like $e^{-\pi T_M/2}$ times a power of T_M, uniformly for $\sigma \leq \operatorname{Re} z \leq \sigma'$, and $\zeta(1+z)$ grows at most polynomially there. The horizontal contributions therefore vanish as $M \to \infty$, the sum of residues converges absolutely, and on the right vertical line $|3^z - 1| \geq 3^{\sigma'} - 1 > 0$ makes $\int_{(\sigma')} |K^*(z)s^{-z}|\,|dz| = O(s^{-\sigma'})$. So

$$K(s) = -\operatorname*{Res}_{z=0}\big[K^*(z)s^{-z}\big] - \sum_{m\neq 0}\operatorname*{Res}_{z=i\omega_m}\big[K^*(z)s^{-z}\big] + O(s^{-\sigma'}).$$

Take the simple poles first. At $z = i\omega_m$ one has $3^{i\omega_m} = e^{2\pi i m} = 1$, so $(3^z - 1)' = 3^z \log 3$ equals c there and $(3/2)^{i\omega_m} = 3^{i\omega_m}2^{-i\omega_m} = 2^{-i\omega_m}$. With $s = e^w$,

$$-\operatorname*{Res}_{z=i\omega_m}\big[K^*(z)e^{-zw}\big] = -\frac{2^{-i\omega_m}}{c}\,\Gamma(i\omega_m)\,\zeta(1+i\omega_m)\,e^{-i\omega_m w},$$

and replacing m by $-m$ throughout the sum turns the whole family into $\hat{H}(m)e^{i\omega_m w}$ with $\hat{H}(m)$ exactly as stated.

At the origin, insert

$$\Gamma(z) = \frac{1}{z} - \gamma_E + \Big(\frac{\gamma_E^2}{2} + \frac{\pi^2}{12}\Big)z + O(z^2), \qquad \zeta(1+z) = \frac{1}{z} + \gamma_E - \gamma_1 z + O(z^2).$$

The two $1/z$ terms cancel against each other and

$$\Gamma(z)\zeta(1+z) = \frac{1}{z^2} + A + O(z), \qquad A = \frac{\pi^2}{12} - \frac{\gamma_E^2}{2} - \gamma_1,$$

which is where A comes from. Also $(3^z - 1)^{-1} = (cz)^{-1} - \frac{1}{2} + cz/12 + O(z^3)$, and $(3/2)^z e^{-zw} = e^{\theta z}$ with $\theta := \log(3/2) - w$. Multiplying the three expansions and reading off the coefficient of z^{-1},

$$\operatorname*{Res}_{z=0}\big[K^*(z)e^{-zw}\big] = \frac{1}{c}\cdot\frac{\theta^2}{2} - \frac{\theta}{2} + \frac{c}{12} + \frac{A}{c}.$$

Substitute $\theta = \log(3/2) - w$ and $\log(3/2) = c - \log 2$. The w-dependent part of $-\operatorname{Res}$ is

$$-\frac{w^2}{2c} + \Big(\frac{\log(3/2)}{c} - \frac{1}{2}\Big)w = -\frac{w^2}{2c} + \Big(\frac{1}{2} - \frac{\log 2}{c}\Big)w = Q(w),$$

and the constant part is $\frac{1}{2}\log 2 - \frac{\log^2 2}{2c} - \frac{c}{12} - \frac{A}{c} = \hat{H}(0)$.

Collecting, with $s = e^w$,

$$L(w) = Q(w) + \widetilde{H}(w) + O(e^{-\sigma' w}), \qquad \widetilde{H}(w) := \hat{H}(0) + \sum_{m\neq 0}\hat{H}(m)e^{i\omega_m w}.$$

That series converges absolutely and uniformly, because $|\Gamma(-i\omega_m)| = \sqrt{\pi/(\omega_m \sinh \pi\omega_m)}$ decays like $e^{-\pi\omega_m/2}$ while $\zeta(1 - i\omega_m)$ grows polynomially, so $\widetilde{H}$ is continuous and exactly c-periodic. It is real valued: Γ and ζ are real on the real axis, so $\hat{H}(-m)$ is the complex conjugate of $\hat{H}(m)$.

The remainder $O(e^{-\sigma' w})$ is only ordinarily exponentially small, weaker than the doubly exponential Δ of Theorem 4, and it is enough. Theorem 4 gives $L - Q - H = \Delta = o(1)$ as $w \to +\infty$, so $\widetilde{H} - H = O(e^{-\sigma' w}) + \Delta(w) \to 0$. Both are c-periodic, and a c-periodic function tending to 0 as $w \to +\infty$ vanishes identically, its value at any w being its value at $w + kc$ for every k. Hence $\widetilde{H} = H$, and by uniform convergence the $\hat{H}(m)$ are the Fourier coefficients of H. □

Remark 7. This is an instance of the de Bruijn–Mahler phenomenon: a base-change harmonic sum, Mellin transformed, picks up simple poles on a vertical line, and their residues assemble into an exactly periodic correction with Fourier coefficients in Γ and ζ. De Bruijn found the first case of this in Mahler's partition asymptotic [5]; Erdős and Richmond name it on p. 448 of [6]; Kato and McLeod [7] and Derfel [8] established the log-squared decay rate of Section 2's comparison asymptotic φ_0 as generic across this whole class of rescaled functional equations. Our contribution here is the closed form for the periodic correction itself, not the mechanism producing it.

The series for H' and H'' converge absolutely and uniformly, since

$$|\Gamma(-i\omega_m)| = \sqrt{\pi/(\omega_m \sinh \pi\omega_m)}$$

decays exponentially in m while $\zeta(1-i\omega_m)$ grows only polynomially: for $\omega \geq \alpha_H := 2\pi/c$ and $N := \lfloor\omega\rfloor \geq 1$, the classical Euler–Maclaurin formula for ζ, $\zeta(s) = \sum_{n=1}^{N} n^{-s} + \frac{N^{1-s}}{s-1} - s\int_N^\infty \{u\}\, u^{-s-1}\, du$ (valid for $\operatorname{Re} s > 0$, $s \neq 1$, integer $N \geq 1$, with $\{u\} := u - \lfloor u\rfloor$), at $s = 1 + i\omega$ gives, bounding each term in modulus ($|N^{-i\omega}| = 1$, $0 \leq \{u\} < 1$),

$$|\zeta(1+i\omega)| \;\leq\; \sum_{n=1}^{N} \frac{1}{n} + \frac{1}{\omega} + \frac{\sqrt{1+\omega^2}}{N} \;\leq\; 1 + \log N + \frac{1}{\alpha_H} + \frac{\sqrt{1+\omega^2}}{N},$$

using $\sum_{n=1}^{N} 1/n \leq 1 + \log N$ and $\omega \geq \alpha_H$ in the last term. Since $N \leq \omega$, $\log N \leq \log\omega$; since $N > \omega - 1$ and $\omega \mapsto \omega/(\omega-1)$ is decreasing for $\omega > 1$, $\sqrt{1+\omega^2}/N < \big(\omega/(\omega-1)\big)\sqrt{1+\omega^{-2}} \leq \big(\alpha_H/(\alpha_H-1)\big)\sqrt{1+\alpha_H^{-2}}$. With

$$C_H := 1 + \alpha_H^{-1} + \frac{\alpha_H}{\alpha_H - 1}\sqrt{1+\alpha_H^{-2}},$$

this gives $|\zeta(1+i\omega)| \leq \log\omega + C_H \leq \log(\omega+1) + C_H$ for $\omega \geq \alpha_H$, and $\zeta(1-i\omega) = \overline{\zeta(1+i\omega)}$ has the same modulus, so $|\zeta(1-i\omega_m)| \leq \log(\omega_m+1) + C_H$. With $q := e^{-\pi\alpha_H/2}$, $A_H := c^{-1}\sqrt{3\pi/\alpha_H}$, the bound $|\Gamma(-i\omega_m)| \leq \sqrt{3\pi/\omega_m}\, e^{-\pi\omega_m/2}$ (valid for $\omega_m \geq \alpha_H$, since $\sinh x \geq e^x/3$ there) together with the bound on $\zeta(1-i\omega_m)$ just proved give the majorant

$$|\hat{H}(m)| \leq A_H\, q^m\, m^{-1/2}\big(\log(\alpha_H m + 1) + C_H\big), \qquad m \geq 1. \tag{4}$$

Since $\sup_w |H^{(k)}(w)| \leq \sum_{m\neq 0} |\omega_m|^k |\hat{H}(m)| = 2\sum_{m\geq 1} \omega_m^k |\hat{H}(m)|$, summing the first six terms with rigorous ball enclosures and bounding the tail $m > 6$ by (4) (the same closed-form geometric-times-polynomial tail sum used in the proof of Proposition 8 below) gives explicit, certified bounds:

$$\sup_w |H'(w)| \leq 0.0011977472315550332, \qquad \sup_w |H''(w)| \leq 0.0068518962896650951. \tag{5}$$

Proposition 8 (Non-constancy and certified oscillation)**.** *H is not constant: $\hat{H}(1) \neq 0$ follows from the classical zero-free theorem for ζ on $\operatorname{Re} s = 1$ alone, with no computation. The oscillation is quantified rigorously by an explicit interval-arithmetic certificate:*

$$-0.000377190280943987 \;<\; H(0) - H(\log(3/2)) \;<\; -0.000377190280943985,$$

and consequently $\operatorname{osc}(H) := \sup H - \inf H \geq 3.7719 \times 10^{-4}$*; evaluating H at two further, well-separated points certifies the stronger bound* $\operatorname{osc}(H) \geq 4.1874494771 \times 10^{-4}$*, and the leading Fourier mode alone certifies a matching upper bound, together giving*

$$4.1874494771 \times 10^{-4} \;\leq\; \operatorname{osc}(H) \;\leq\; 4.187981 \times 10^{-4}.$$

Proof. Non-constancy first, unconditionally. By Proposition 6, $\hat{H}(1) = -\frac{2^{i\omega_1}}{c}\Gamma(-i\omega_1)\zeta(1-i\omega_1)$, $\omega_1 = 2\pi/c$, a product of three factors, each nonzero: $2^{i\omega_1} \neq 0$ trivially; Γ has no zeros anywhere, and $-i\omega_1$ is not one of its poles ($\omega_1 \neq 0$ is real, and Γ's poles are only at the non-positive integers), so $\Gamma(-i\omega_1) \neq 0$; and $1 - i\omega_1$ lies on the line $\operatorname{Re} s = 1$ with $\omega_1 \neq 0$, where $\zeta(1+it) \neq 0$ for every real $t \neq 0$ by the classical zero-free theorem of Hadamard and de la Vallée Poussin [9], so $\zeta(1-i\omega_1) \neq 0$. Hence $\hat{H}(1) \neq 0$, and a periodic function with a nonzero Fourier coefficient at $m \neq 0$ cannot be constant.

For the quantitative bound, write $g(b) = -b/2 + \log S(b/2)$, $S(x) = \sinh(x)/x = \sum_{n\geq 0} x^{2n}/(2n+1)!$ (this is g's defining formula rewritten around its removable singularity at 0, since $e^{-b/2}(e^{b/2} - e^{-b/2}) = 1 - e^{-b}$). Termwise domination of $6^n n! \leq (2n+1)!$, which holds at $n = 0$ and propagates because $(2n+3)(2n+2) \geq 6(n+1)$, gives $1 \leq S(x) \leq e^{x^2/6}$, that is

$$0 \ \leq\ \log S(x) \ \leq\ \frac{x^2}{6}, \qquad x \geq 0. \tag{6}$$

Since $\sum_{j\geq 1} 3^{-j} = \frac{1}{2}$, the $-b/2$ halves of the terms of L sum in closed form, and (2) becomes, with $s = e^w$,

$$H(w) = -\frac{s}{2} + \sum_{j\geq 1} \log S\Big(\frac{s}{3^j}\Big) - Q(w) + \sum_{k\geq 0} \log\big(1 - e^{-2\cdot 3^k s}\big). \tag{7}$$

Truncate the first series at $j = R$ and the second at $k = M$. By (6) the first tail satisfies

$$0 \ \leq\ \sum_{j>R} \log S\Big(\frac{s}{3^j}\Big) \ \leq\ \frac{s^2}{6} \sum_{j>R} 9^{-j} = \frac{s^2\, 9^{-R}}{48},$$

and, since $0 \leq -\log(1-y) \leq y/(1-y)$ and $e^{-2\cdot 3^k s} = \big(e^{-2\cdot 3^{M+1} s}\big)^{3^{k-M-1}}$ for $k > M$, the second tail satisfies

$$0 \ \leq\ -\sum_{k>M} \log\big(1 - e^{-2\cdot 3^k s}\big) \ \leq\ \frac{y}{(1-y)^2} \ \leq\ 4y, \qquad y := e^{-2\cdot 3^{M+1} s} \leq \tfrac{1}{2}.$$

Take $R = 30$, $M = 4$. For $s \leq 3$ (which covers every evaluation point used below, including the oscillation grid) the first tail is at most 4.43×10^{-30}, and for $s \geq 1$ the second is at most $4e^{-486} < 10^{-210}$. The remaining 35 terms are elementary functions of s; evaluated at 50 significant decimal digits, where the accumulated rounding stays below 10^{-45}, they give

$$H(0) = -0.62713093305152597830\ldots, \qquad H\big(\log \tfrac{3}{2}\big) = -0.62675374277058199247\ldots,$$

whose difference is $-0.00037719028094398583\ldots$, inside the stated interval; this first route is floating-point (50 significant digits, accumulated rounding bounded as stated) rather than interval arithmetic. The Fourier series of Proposition 6 gives the same difference in Arb ball arithmetic, rigorously: the $m = 0$ term cancels, and at 250-bit working precision (ball radius $< 10^{-70}$ throughout) the four nonzero modes needed are

$$\begin{aligned}
\hat{H}(1) &= -1.020544273068431785 \times 10^{-4} - 2.332597606471452884 \times 10^{-5}\, i,\\
\hat{H}(2) &= -6.523276079851823231 \times 10^{-9} + 1.156720838500605667 \times 10^{-8}\, i,\\
\hat{H}(3) &= -1.799881020474424677 \times 10^{-12} + 6.648927223521397737 \times 10^{-13}\, i,\\
\hat{H}(4) &= -9.219524659678388767 \times 10^{-17} + 9.444517590655925766 \times 10^{-17}\, i,
\end{aligned}$$

the four-mode sum

$$2\,\mathrm{Re} \sum_{m=1}^{4} \hat{H}(m)\big(1 - e^{i\omega_m \log(3/2)}\big) = -0.0003771902809439858148\ldots$$

(ball radius $< 10^{-70}$), matching the floating-point value above to 19 decimal places (16 significant digits). The majorant (4) gives $\sum_{m>4} |\hat{H}(m)| \leq 9.41 \times 10^{-20}$, so the discarded modes move the four-mode sum by less than 3.8×10^{-19}, giving

$$H(0) - H\big(\log \tfrac{3}{2}\big) \in \big[-0.0003771902809439861948,\ -0.0003771902809439854348\big];$$

this route alone certifies the stated interval for $H(0) - H(\log(3/2))$.

For the stronger bound, let $w_i = ic/N$, $0 \leq i < N$, with $N = 2^{20}$. A floating-point evaluation of the N grid values from Proposition 6, truncated at $|m| \leq 10$ (the same majorant bounds each pointwise truncation error by $2 \times 2.80 \times 10^{-43} = 5.60 \times 10^{-43}$, combining the discarded modes m and $-m$), locates a candidate maximum at $i = 486746$ and a candidate minimum at $i = 1011118$; this search is only a heuristic for finding two well-separated points and is not itself part of the certificate below. Re-evaluating H at exactly these two points at 250-bit working precision in Arb ball arithmetic, independently, from (7) and from the Fourier series (the two routes agree to about 30 digits, consistent with (7)'s own $R = 30$ truncation tail bound above; a cross-check, not part of the certificate),

$$H(w_{486746}) = -0.6267174403736425786\ldots, \qquad H(w_{1011118}) = -0.6271361853213578093\ldots,$$

so $D := H(w_{486746}) - H(w_{1011118}) = 4.187449477152 \times 10^{-4}$. The certified enclosure uses only the Fourier route: the Arb ball rounding at 250-bit precision is $< 10^{-70}$ per evaluation, negligible next to the majorant truncation error already bounded above, 5.60×10^{-43} per point, giving D to radius $< 1.2 \times 10^{-42}$ (twice that per-point bound). Since w_{486746} and $w_{1011118}$ are two specific, fixed points of the domain, $H(w_{486746}) \leq \sup H$ and $H(w_{1011118}) \geq \inf H$ hold trivially, giving $D \leq \operatorname{osc}(H)$: this bound is unconditional, independent of whether the floating-point search above actually located the true argmax and argmin, since it uses only that the two chosen points lie in the domain.

For the matching upper bound, no search is needed. Write $H = \hat{H}(0) + 2\operatorname{Re}\big(\hat{H}(1)e^{i\omega_1 w}\big) + \Xi(w)$, $\Xi(w) := 2\sum_{m\geq 2}\operatorname{Re}\big(\hat{H}(m)e^{i\omega_m w}\big)$, so $|\Xi(w)| \leq 2\sum_{m\geq 2}|\hat{H}(m)|$ for every w. The one-mode term $2\operatorname{Re}(\hat{H}(1)e^{i\omega_1 w})$ ranges, as w varies, over an interval of width exactly $4|\hat{H}(1)|$, so $\operatorname{osc}(H) \leq 4|\hat{H}(1)| + 4\sum_{m\geq 2}|\hat{H}(m)|$. Using the already-certified $\hat{H}(2)$, $\hat{H}(3)$, $\hat{H}(4)$ from above and the majorant tail $\sum_{m>4}|\hat{H}(m)| \leq 9.41 \times 10^{-20}$,

$$\sum_{m\geq 2}|\hat{H}(m)| \leq |\hat{H}(2)| + |\hat{H}(3)| + |\hat{H}(4)| + 9.41 \times 10^{-20} \leq 1.3282 \times 10^{-8},$$

and $4|\hat{H}(1)| \leq 4.187449304 \times 10^{-4}$, giving $\operatorname{osc}(H) \leq 4.187981 \times 10^{-4}$, rigorously and unconditionally: this bound uses only the already-certified Fourier coefficients, not the grid search. □

4. A uniform saddlepoint approximation

Define, for $s > 0$,

$$\varphi_{\mathrm{sp}}(t(s)) := \frac{\exp\big(K(s) + st(s)\big)}{\sqrt{2\pi K''(s)}}.$$

This section proves that $\varphi(t(s))/\varphi_{\mathrm{sp}}(t(s)) \to 1$ as $s \to \infty$, uniformly over $\rho := 2s/3^N \in [1, 3)$ (defined below; this parameter records the same phase, $w \bmod c$, tracked through H in Sections 6 and 7). That uniformity is what lets the two theorems of Section 7 depend on different scopes rather than on different mathematics.

For $n \geq 2$, define the derivative-normalized functions $\kappa_n(w) = (-1)^n w^n g^{(n)}(w)$ on $\{\operatorname{Re} w > 0\}$; $\kappa_n(b)$, $b > 0$, is b^n times the n-th cumulant of the random variable with density $b\,e^{-bv}/(1 - e^{-b})$ on $[0, 1]$. Writing $x = w/2$,

$$\kappa_2(w) = 1 - \frac{x^2}{\sinh^2 x}, \qquad \kappa_3(w) = 2 - \frac{2x^3\cosh x}{\sinh^3 x}, \qquad \kappa_4(w) = 6 + \frac{2x^4(1 - 3\coth^2 x)}{\sinh^2 x},$$

and from the Bernoulli expansion $g(w) + w/2 = \sum_{k\geq 1} B_{2k}w^{2k}/(2k(2k)!)$ on $|w| < 2\pi$,

$$\kappa_n(w) = (-1)^n \sum_{k\geq\lceil n/2\rceil} \frac{B_{2k}\,w^{2k}}{2k\,(2k-n)!}, \qquad n \geq 2,$$

so that $\kappa_n(w) = O(w^2)$ as $w \to 0$ for every $n \geq 2$.

Write $s = \rho \cdot 3^N/2$ for the unique $N = \lfloor \log_3(2s) \rfloor \in \mathbb{Z}$ and $\rho \in [1, 3)$, and $b_j = 2s/3^j$. Then $\{b_j : j \geq 1\} = \{\rho \cdot 3^m : m \leq N - 1\}$; for $N \geq 1$ (the only range used from here on, corresponding to $s \geq 3/2$), exactly N of these, the smallest being $\rho \geq 1$, are ≥ 1, and the rest form a geometric tail below 1. This single structural fact is the source of every phase-independent constant below.

Lemma 9 (Boundedness off the real axis)**.** *For $w = b(1+iu)$, $b > 0$, $|u| \leq 1$, with $e(r) = r^2/\sinh^2 r$ and $f(r) = r^3 \cosh r/\sinh^3 r$,*

$$|\kappa_2(w)| \leq 1 + 2e(b/2), \qquad |\kappa_3(w)| \leq 2 + 2 \cdot 2^{3/2} f(b/2).$$

Both bounds are finite over the whole sector $|\operatorname{Im} w| \leq \operatorname{Re} w$: $\sup|\kappa_2| \leq 3$, $\sup|\kappa_3| \leq 2+2^{5/2} < 7.657$. In addition, for $|w| \leq 2$, $|\kappa_2(w)| \leq 0.114|w|^2$ and $|\kappa_3(w)| \leq 0.0119|w|^4$.

Proof. $|\sinh z|^2 = \sinh^2(\operatorname{Re} z) + \sin^2(\operatorname{Im} z) \geq \sinh^2(\operatorname{Re} z)$ and $|\cosh z|^2 \leq \cosh^2(\operatorname{Re} z)$; the first gives $|\sinh(w/2)| \geq \sinh(b/2) > 0$, excluding the poles of coth, since $\operatorname{Re}(w/2) = b/2 > 0$. This gives the sector bounds directly. Monotonicity of e is immediate from the increasing power series of $\sinh(r)/r$. For f, set $u = 2r$; a direct computation gives $2r\sinh(r)\cosh(r)\,(\log f)'(r) = F(u)$ with $F(u) := 3\sinh u - u\cosh u - 2u$, so $(\log f)'(r)$ has the sign of $F(2r)$. Since $F(0) = F'(0) = F''(0) = 0$ and $F'''(u) = -u\sinh u < 0$ for $u > 0$, integrating three times from 0 gives $F(u) < 0$ for all $u > 0$, hence $(\log f)'(r) < 0$ for $r > 0$ and f is strictly decreasing. (Lazarević's inequality, $(\sinh r/r)^3 > \cosh r$, separately gives $f < 1$.) The local bounds at $|w| \leq 2$ follow by applying the maximum-modulus principle to $\kappa_2(w)/w^2$ and $\kappa_3(w)/w^4$ (analytic on $|w| < 2\pi$ by the Bernoulli expansion): bound each on $|w| = 2$ by the triangle inequality, using $|B_{2k}| = 2(2k)!\,\zeta(2k)/(2\pi)^{2k}$ and $\zeta(2k) \leq \zeta(2)$, respectively $\zeta(4)$. ☐

Lemma 10 (Variance)**.** *For all $\rho \in [1,3)$, $N \geq 1$, with $V := s^2 K''(s) = \sum_j \kappa_2(b_j)$,*

$$N - A_V \leq V \leq N + 0.1283, \qquad A_V := \sum_{m \geq 0} e(3^m/2) = 1.4269413069\ldots.$$

Proof. Split $V = \sum_{m=0}^{N-1} \kappa_2(\rho 3^m) + \sum_{m<0} \kappa_2(\rho 3^m)$, using $\kappa_2(b) = 1 - e(b/2) \in (0,1)$. For the lower bound, drop the second (non-negative) sum and use $e(\rho 3^m/2) \leq e(3^m/2)$ (e decreasing, $\rho \geq 1$) in the first, giving $\sum_{m=0}^{N-1} \kappa_2(\rho 3^m) \geq N - A_V$. For the upper bound, the first sum is $< N$ term by term, and the second sum, where $\rho 3^m < 1$, is bounded by Lemma 9's local estimate: $\sum_{m<0} \kappa_2(\rho 3^m) \leq 0.114\rho^2 \sum_{m<0} 9^m = 0.114\rho^2/8 \leq 0.1283$ for $\rho < 3$. ☐

Lemma 11 (Tail bound and existence of the density)**.** *If $b_0 \geq 1$ and $b_k = 3^k b_0$, then*

$$\sum_{k \geq 0} \log\coth(b_k/2) \leq 3e^{-b_0}.$$

Consequently, with $\Lambda(y) := K(s(1+iy)) - K(s) + isy\,t(s)$,

$$|e^{\Lambda(y)}| \leq e^{3/e}(1+y^2)^{-N/2} \leq 3.0152\,(1+y^2)^{-N/2}$$

for every real y. In particular X has a continuous density, the tilted characteristic function is integrable for $N \geq 3$, and Fourier inversion at $x = t(s)$ gives

$$R(s) := \frac{\varphi(t(s))}{\varphi_{\mathrm{sp}}(t(s))} = \sqrt{\frac{V}{2\pi}} \int_{\mathbb{R}} e^{\Lambda(y)}\,dy, \tag{8}$$

a real quantity, since $\Lambda(-y) = \overline{\Lambda(y)}$.

Proof. $\log\coth(x/2) = 2\operatorname{artanh}(e^{-x}) \leq 2e^{-x}/(1-e^{-2x})$; summing over the geometric orbit $b_k = 3^k b_0$ and bounding the resulting series by its first few terms gives the stated constant. X's density is continuous because the sum of the first two summands already has a continuous, compactly supported density and convolution preserves continuity. For Λ: $t(s) = -K'(s)$ cancels the linear term of the tilted transform exactly, and each remaining factor $e^{g(b_j(1+iy))-g(b_j)}$ has modulus at most 1 (a tilted characteristic function), with modulus at most $\coth(b_j/2)/\sqrt{1+y^2}$ on the N indices where $b_j \geq 1$; the tail estimate applies to exactly that geometric set, with $b_0 = \rho \geq 1$. ☐

Lemma 12 (Central expansion)**.** *With $h(u) = g(b(1+iu))$, Taylor's theorem in* integral *form (the Lagrange form is invalid for a complex-valued function of a real variable) gives*

$$h(y) = h(0) + h'(0)y + h''(0)\tfrac{y^2}{2} + \tfrac{1}{2}\int_0^y (y-u)^2 h'''(u)\,du.$$

Consequently, writing $h_j(u) := g(b_j(1+iu))$ for each $j \ge 1$, for $\theta \in (0,1]$,

$$S := \sum_j \sup_{|u|\le\theta} |h_j'''(u)| \le 2N + 10.559,$$

and hence $|\Lambda(y) + Vy^2/2| \le B|y|^3$ for $|y| \le \theta$, with $B := S/6 \le (2N + 10.559)/6$.

Proof. Write $w_j := b_j(1+iu)$. The chain rule and the definition of κ_3 give $h_j'''(u) = i\,\kappa_3(w_j)/(1+iu)^3$, so

$$|h_j'''(u)| = \frac{|\kappa_3(w_j)|}{(1+u^2)^{3/2}}.$$

Split the index set as in Section 4. For the N indices with $b_j = \rho 3^m \ge 1$, $0 \le m \le N-1$, drop the denominator and use Lemma 9's sector bound, legitimate since $|u| \le \theta \le 1$:

$$|h_j'''(u)| \le |\kappa_3(w_j)| \le 2 + 2^{5/2} f(b_j/2) \le 2 + 2^{5/2} f(3^m/2),$$

the last step by monotonicity of f and $\rho \ge 1$. Summing over $m \ge 0$ bounds this block by $2N + 2^{5/2}\Sigma$, $\Sigma := \sum_{m\ge0} f(3^m/2)$. The first three terms of Σ are $0.99614738\ldots$, $0.82241066\ldots$, $0.04500508\ldots$; for $r \ge 13$ one has $f(r) \le 4.001\, r^3 e^{-2r}$, so the remaining terms sum to less than 2×10^{-8}, and $2^{5/2}\Sigma < 10.541906$.

For the remaining indices $b_j = \rho 3^{-n}$, $n \ge 1$, one has $b_j < 1$ and $|w_j| = b_j\sqrt{1+u^2} \le b_j\sqrt{2} < 2$, so Lemma 9's local bound applies and, keeping the denominator this time,

$$|h_j'''(u)| \le \frac{0.0119\,|w_j|^4}{(1+u^2)^{3/2}} = 0.0119\, b_j^4\,(1+u^2)^{1/2} \le 0.0119\sqrt{2}\, b_j^4.$$

Summing, $\sum_{n\ge1}(\rho 3^{-n})^4 = \rho^4/80 < 81/80$, so this block is below $0.0119\sqrt{2}\cdot 81/80 < 0.01704$. Together, $S < 2N + 10.541906 + 0.01704 < 2N + 10.559$.

For the remainder bound, $\sum_j h_j(y) = K(s(1+iy))$, $\sum_j h_j'(0) = isK'(s) = -is\,t(s)$ and $\sum_j h_j''(0) = -s^2K''(s) = -V$, so summing the integral-form Taylor expansion over j gives

$$\Lambda(y) + \frac{Vy^2}{2} = \sum_j \frac{1}{2}\int_0^y (y-u)^2 h_j'''(u)\,du,$$

whose modulus is at most $S|y|^3/6 = B|y|^3$ for $|y| \le \theta$. □

Theorem 13 (Uniform saddlepoint asymptotic)**.** *For $N \ge 19$, uniformly over $\rho \in [1,3)$, $|R(s) - 1| \le E(N) := e_1(N) + e_2(N) + e_3(N)$, with $\theta_N := N^{-1/3}$, $V_{\rm lo} := N - A_V$, $V_{\rm up} := N + 0.1283$, $B := (2N + 10.559)/6$,*

$$e_1(N) := \frac{4\,e^{B\theta_N^3}\,B}{\sqrt{2\pi}\,V_{\rm lo}^{3/2}}, \qquad e_2(N) := \frac{2}{\sqrt{2\pi}}\cdot\frac{e^{-\theta_N^2 V_{\rm lo}/2}}{\theta_N\sqrt{V_{\rm lo}}}, \tag{9}$$

$$e_3(N) := 2\,e^{3/e}\sqrt{\frac{V_{\rm up}}{2\pi}}\cdot\frac{(1+\theta_N^2)^{-(N-2)/2}}{\theta_N(N-2)}.$$

$\sqrt{N}\,E(N) \to \frac{4}{3}e^{1/3}(2\pi)^{-1/2} = 0.742358\ldots$ as $N \to \infty$ (proved below, from e_1 alone: e_2 and e_3 vanish faster than any power of N). Consequently $\varphi(t) = \varphi_{\rm sp}(t)\,(1 + o(1))$ as $t \to 0^+$, with no restriction on the sequence of t's.

Remark 14. The estimate $|R(s)-1| \le E(N)$ becomes informative once $E(N) < 1$, which holds at $N = 19$ (see the proof: $E(18) = 1.00170\ldots \ge 1$, $E(19) = 0.95673\ldots < 1$); this is the threshold kept as the theorem's hypothesis (whether some smaller N also gives $E(N) < 1$ is immaterial to what follows and is not checked). E is also strictly decreasing and satisfies $E(N) \le 4.18\, N^{-1/2}$ throughout $19 \le N \le 5000$ (checked directly in the accompanying repository), though only $E(N) \to 0$ as $N \to \infty$ is used below.

Proof. Split (8) at $|y| = \theta_N$. On $|y| \le \theta_N$, Lemma 12 bounds the cubic remainder $|\Lambda(y) + Vy^2/2|$ by $B\theta_N^3$, which stays bounded (not $o(1)$: this is exactly why e_1 carries the factor $e^{B\theta_N^3}$ rather than vanishing) as $N \to \infty$, since $B = O(N)$ and $\theta_N^3 = N^{-1}$. Write $C(y) := e^{\Lambda(y)} - e^{-Vy^2/2}$ and factor the Gaussian out before estimating, $e^{\Lambda(y)} = e^{-Vy^2/2}\, e^{z(y)}$ with $z(y) := \Lambda(y) + Vy^2/2$, so that $C(y) = e^{-Vy^2/2}\big(e^{z(y)} - 1\big)$. Since $|z(y)| \le B|y|^3 \le B\theta_N^3$ on $|y| \le \theta_N$, the elementary $|e^z - 1| \le |z|e^{|z|}$ gives

$$|C(y)| \;\le\; e^{-Vy^2/2}\, B|y|^3\, e^{B\theta_N^3} \qquad (|y| \le \theta_N),$$

the Gaussian factor retained. Integrating that inequality, $\int_{\mathbb{R}} e^{-Vy^2/2}|y|^3\,dy = 4/V^2$, so

$$\sqrt{V/2\pi}\int_{|y|\le\theta_N} |C(y)|\,dy \;\le\; \frac{4Be^{B\theta_N^3}}{\sqrt{2\pi}\,V^{3/2}},$$

which is decreasing in V; using $V \ge V_{\mathrm{lo}}$ from Lemma 10 gives the central Taylor-remainder contribution e_1. Since $\sqrt{V/2\pi}\int_{\mathbb{R}} e^{-Vy^2/2}\,dy = 1$, the same splitting leaves exactly two further terms. The truncated Gaussian tail $\sqrt{V/2\pi}\cdot 2\int_{\theta_N}^{\infty} e^{-Vy^2/2}\,dy$, substituting $v = y\sqrt{V}$ to get $(2/\sqrt{2\pi})\int_{c_0}^{\infty} e^{-v^2/2}\,dv$, $c_0 = \theta_N\sqrt{V}$ (a local constant, distinct from $c := \log 3$), and bounding via $\int_{c_0}^{\infty} e^{-v^2/2}\,dv \le e^{-c_0^2/2}/c_0$ at $V = V_{\mathrm{lo}}$, gives e_2. On $|y| > \theta_N$, Lemma 11's bound $|e^{\Lambda(y)}| \le e^{3/e}(1+y^2)^{-N/2}$ integrates, via $\int_u^{\infty}(1+y^2)^{-N/2}\,dy \le (1+u^2)^{-(N-2)/2}/(u(N-2))$ at $u = \theta_N$ (using $V \le V_{\mathrm{up}}$ for the leading $\sqrt{V/2\pi}$ factor in (8)), to give e_3; this needs $\theta_N \gg N^{-1/2}$, satisfied since $\frac{1}{3} < \frac{1}{2}$. Every constant entering E (via A_V, Lemma 10, and the $3.0152 \ge e^{3/e}$ of Lemma 11) is a supremum over $b > 0$ or a sum over one complete geometric orbit bounded using $\rho \ge 1$, hence independent of ρ. Directly evaluating (9) gives $E(18) = 1.00170\ldots \ge 1$ and $E(19) = 0.95673\ldots < 1$, matching the remark above.

For the asymptotic rate, e_1, e_2, e_3 separate on very different scales as $N \to \infty$. Since $\theta_N^3 = 1/N$, $B\theta_N^3 \to 1/3$ and $B/N \to 1/3$, V_{lo}/N, $V_{\mathrm{up}}/N \to 1$, so

$$\sqrt{N}\,e_1(N) = \frac{4e^{B\theta_N^3}}{\sqrt{2\pi}}\cdot\frac{\sqrt{N}\,B}{V_{\mathrm{lo}}^{3/2}} \longrightarrow \frac{4e^{1/3}}{\sqrt{2\pi}}\cdot\frac{1}{3} = \frac{4}{3}e^{1/3}(2\pi)^{-1/2}.$$

For e_2 and e_3, $\theta_N^2 V_{\mathrm{lo}} \sim N^{1/3} \to \infty$, so both carry a factor $e^{-N^{1/3}/2}(1+o(1))$ (for e_3, via $(1+\theta_N^2)^{-(N-2)/2} = \exp(-\frac{N-2}{2}\theta_N^2(1+o(1)))$), against polynomial prefactors of order $N^{-1/6}$; a factor decaying like $e^{-N^{1/3}/2}$ beats every power of N, so $N^k e_2(N)$, $N^k e_3(N) \to 0$ for every k. Hence $\sqrt{N}\,E(N) = \sqrt{N}\,e_1(N) + o(1) \to \frac{4}{3}e^{1/3}(2\pi)^{-1/2}$, and in particular $E(N) \to 0$. Since $t(s)$ ranges bijectively over $(0, 1/2)$, this is exactly the statement $\varphi(t) = \varphi_{\mathrm{sp}}(t)(1+o(1))$ as $t \to 0^+$. ☐

Remark 15. The rate is not sharp, and a sharper one is plausible but not established here. Carrying the central expansion to fourth order, using that the odd cubic term integrates to zero against the Gaussian core, and using the limits $V_3 := \sum_j \kappa_3(b_j) = 2N + O(1)$, $V_4 := \sum_j \kappa_4(b_j) = 6N + O(1)$ (the third and fourth cumulant sums, by the same argument as Lemma 10) suggests the Edgeworth correction $R(s) = 1 - \frac{1}{12N} + O(N^{-3/2})$, matching the leading term of the classical Stirling correction for a Gamma$(N,1)$ law: after tilting, the $b_j \ge 1$ summands behave asymptotically like independent Exp(1) variables, so X under its tilt is asymptotically Gamma$(N,1)$-distributed, and ρ enters only through $O(1)$ shifts of the cumulants, at order N^{-2}. We do not need this sharper rate and do not prove the remainder term here; Theorem 13's qualitative $o(1)$ is all that Section 7 uses.

5. The envelope lemma

Theorem 13 compares φ to φ_{sp}, a quantity built from the full transform $K = Q + H + \Delta$. It remains to compare φ_{sp} to a smooth auxiliary system built from Q alone, so that the periodic part H can be extracted explicitly. Write $t = e^{-\tau}$, $g_\tau(w) = L(w) + e^{w-\tau}$, $g_{0,\tau}(w) = Q(w) + e^{w-\tau}$. The true and smooth systems each have their own analogue of $-K'$: write $B_{\mathrm{tr}}(w) := e^w t(e^w) = -L'(w)$ and $B_{\mathrm{sm}}(w) := -Q'(w) = w/c - a$.

Lemma 16 (True and smooth saddles)**.** *Let $\Phi(w) = w - \log B_{\mathrm{tr}}(w)$ and $\Phi_0(w) = w - \log B_{\mathrm{sm}}(w)$, and set $\tau_0 := 1 + ca + \log c = 0.9502067913\ldots$, so $\tau_0 > \log 2$.*

For $\tau > \log 2$, g_τ has a unique global minimizer $w^*(\tau)$, at which $g_\tau''(w^*) = V(r^*) > 0$, $r^* = e^{w^*}$; it is characterized by $\tau = \Phi(w^*)$, and Φ is a strictly increasing bijection $\mathbb{R} \to (\log 2, \infty)$. For $\tau \le \log 2$ there is no critical point at all. For $\tau > \tau_0$, $g_{0,\tau}$ has a unique local minimizer $w_0(\tau)$, characterized by $\tau = \Phi_0(w_0)$ together with $B_0 := B_{\mathrm{sm}}(w_0) = e^{w_0-\tau} > 1/c$; Φ_0 restricted to $(ca+1, \infty)$ is a strictly increasing bijection onto (τ_0, ∞), and B_0 increases from $1/c$ to ∞ along it. For $\tau \le \tau_0$ there is no such minimizer.

Once $B_0 \ge 5$ (equivalently $\tau \ge 3.73978\ldots$), $|w^* - w_0| \le 0.00652/B_0$. Moreover $w_0(\tau + c) - w_0(\tau) = c + O(1/\tau)$ as $\tau \to \infty$.

Proof. Substituting $s = e^w$ and $t = e^{-\tau}$ turns g_τ into $K(s) + st$, strictly convex in $s > 0$ because $K'' > 0$, with derivative $K'(s) + t$ vanishing exactly where $t = t(s)$. As $t(\cdot)$ decreases strictly from $1/2$ to 0, that happens for exactly one s when $t \in (0, 1/2)$, that is when $\tau > \log 2$, and for no s when $t \ge 1/2$. Since $w \mapsto e^w$ is a bijection $\mathbb{R} \to (0,\infty)$, $w^* = \log s$ is the unique global minimizer of g_τ, and Φ inverts $\tau \mapsto w^*$. Differentiating twice, $g_\tau''(w) = e^w\big(K'(e^w) + t\big) + e^{2w}K''(e^w)$, whose first term vanishes at w^*, leaving $g_\tau''(w^*) = V(r^*) > 0$.

For the smooth system, $g_{0,\tau}'(w) = e^{w-\tau} - B_{\mathrm{sm}}(w)$ and $g_{0,\tau}''(w) = e^{w-\tau} - 1/c$, so a critical point is a local minimizer exactly when $B_0 > 1/c$. Also $\Phi_0'(w) = 1 - 1/(c\,B_{\mathrm{sm}}(w))$, positive exactly where $B_{\mathrm{sm}}(w) > 1/c$, that is on $w > ca + 1$; there Φ_0 increases from $\Phi_0(ca+1) = ca + 1 + \log c = \tau_0$ to ∞. This gives the stated existence, uniqueness and range.

Now take $B_0 \ge 5$. The decomposition $L = Q + H + \Delta$ evaluates the true gradient at the smooth saddle exactly: $Q'(w_0) = -B_0 = -e^{w_0-\tau}$, so the leading terms cancel and

$$g_\tau'(w_0) = L'(w_0) + e^{w_0-\tau} = H'(w_0) + \Delta'(w_0), \qquad |g_\tau'(w_0)| \le \varepsilon_1 := \sup|H'| + \sup_{[w_0-1,w_0+1]} |\Delta'|. \tag{10}$$

On $[w_0 - 1, w_0 + 1]$ write $g_\tau''(w) = e^{w-\tau} - 1/c + H''(w) + \Delta''(w)$. Here $e^{w-\tau} = B_0 e^{w-w_0}$ ranges over $[B_0/e,\ B_0 e]$, so g_τ'' admits no bound independent of τ; it is of exact order B_0,

$$\frac{B_0}{e} - \eta_0 \ \le\ g_\tau''(w) \ \le\ B_0 e, \qquad \eta_0 := \frac{1}{c} + \sup|H''| + \sup_{[w_0-1,w_0+1]} |\Delta''|, \tag{11}$$

the upper bound because $-1/c + \sup|H''| + \sup|\Delta''| < 0$. Since $w_0 = c(B_0 + a)$, $B_0 \ge 5$ forces $w_0 \ge 5.3492$ and so $w \ge 4.3492$ on the interval, where $|\Delta'|, |\Delta''| \le 10^{-60}$: with $b := 2e^w \ge 2e^{4.3492} > 154$ and $x_k := 3^k b$, differentiating $\Delta(w) = -\sum_{k\ge0} \log(1 - e^{-x_k})$ termwise (justified as in the proof of Theorem 4) gives $\Delta'(w) = -\sum_{k\ge0} x_k/(e^{x_k} - 1)$ and, applying $|d/dx[x/(e^x-1)]| \le 8xe^{-x}$ for $x \ge 1$ (elementary, since $e^x - 1 \ge e^x/2$ there) together with $dx_k/dw = x_k$, $\Delta''(w) = -\sum_{k\ge0} x_k \cdot \frac{d}{dx}\left[\frac{x}{e^x-1}\right]_{x=x_k}$, so $|\Delta''(w)| \le \sum_{k\ge0} 8x_k^2 e^{-x_k}$; bounding both sums by their dominant $k = 0$ term via $x_k - b \ge 2kb$ (from $3^k - 1 \ge 2k$) gives $|\Delta'(w)| \le 4be^{-b}$ and $|\Delta''(w)| \le 16b^2e^{-b}$, both well under 10^{-60} at $b > 154$. With (5) this makes $\varepsilon_1 \le 0.0011978$ and $\eta_0 \le 0.9170912$, whence

$2e\eta_0 \le 4.9859 \le 5 \le B_0$, that is $\eta_0 \le B_0/(2e)$, and (11) improves to $g''_\tau \ge B_0/(2e) > 0$ throughout the interval.

Put $h_0 := 2e\varepsilon_1/B_0 \le 2e \cdot 0.0011978/5 < 0.0014$. For $h \in (h_0, 1]$ the mean value theorem applied on $[w_0-1, w_0+1]$ gives $g'_\tau(w_0+h) \ge g'_\tau(w_0) + hB_0/(2e) \ge -\varepsilon_1 + hB_0/(2e) > 0$ and, symmetrically, $g'_\tau(w_0-h) \le \varepsilon_1 - hB_0/(2e) < 0$. As w^* is the only zero of g'_τ, it lies in (w_0-h, w_0+h) for every such h, so $|w^*-w_0| \le h_0 = 2e\varepsilon_1/B_0 \le 0.00652/B_0$.

For the phase drift, $w_0 = c(B_0+a)$ and $\tau = w_0 - \log B_0$ give $dw_0/d\tau = (1-1/(cB_0))^{-1} = 1 + O(1/B_0)$ along the curve $\tau \mapsto w_0(\tau)$; since $B_0 \to \infty$ as $\tau \to \infty$ (via $B_0 \sim \tau/c$), integrating over one period gives $w_0(\tau+c) - w_0(\tau) = \int_\tau^{\tau+c}(1+O(1/B_0(u)))\,du = c + O(1/\tau)$. ☐

Proposition 17 (Envelope estimate)**.** *With $S(\tau) := \log\varphi_{\mathrm{sp}}(t) = g_\tau(w^*) + w^* - \frac{1}{2}\log(2\pi V(r^*))$ and $P(\tau) := Q(w_0) + B_0 + w_0 - \frac{1}{2}\log\bigl(2\pi(B_0 - \frac{1}{c})\bigr)$,*

$$|S(\tau) - P(\tau) - H(w_0(\tau))| = O(1/\tau) \qquad (\tau \to \infty),$$

uniformly in the phase of τ modulo c.

Proof. Start from the identity. Since $g_{0,\tau}(w_0) = Q(w_0) + e^{w_0-\tau} = Q(w_0) + B_0$, the definition of P reads $P(\tau) = g_{0,\tau}(w_0) + w_0 - \frac{1}{2}\log\bigl(2\pi(B_0 - \frac{1}{c})\bigr)$, so

$$S(\tau) - P(\tau) = \bigl[g_\tau(w^*) - g_{0,\tau}(w_0)\bigr] + (w^* - w_0) - \frac{1}{2}\log\frac{V(r^*)}{B_0 - \frac{1}{c}}.$$

Theorem 4 gives $L = Q + H + \Delta$, hence $g_\tau = g_{0,\tau} + H + \Delta$ pointwise, and in particular $g_\tau(w^*) = g_{0,\tau}(w^*) + H(w^*) + \Delta(w^*)$. Substituting and subtracting $H(w_0)$,

$$\begin{aligned} S(\tau) - P(\tau) - H(w_0) = {} & \underbrace{\bigl[g_{0,\tau}(w^*) - g_{0,\tau}(w_0)\bigr]}_{T_1} + \underbrace{\bigl[H(w^*) - H(w_0)\bigr]}_{T_2} + \underbrace{(w^* - w_0)}_{T_3} \\ & \underbrace{- \frac{1}{2}\log\frac{V(r^*)}{B_0 - \frac{1}{c}}}_{T_4} + \underbrace{\Delta(w^*)}_{T_5}. \end{aligned} \tag{12}$$

This is exact for every τ with $B_0 \ge 5$. The five terms are bounded in turn.

T_1 is the envelope gap of the smooth system. Taylor about w_0, where $g'_{0,\tau}$ vanishes, with $g''_{0,\tau}(w) = e^{w-\tau} - 1/c \le B_0 e$ on $[w_0-1, w_0+1]$ and $|w^* - w_0| \le 0.00652/B_0$ from Lemma 16, gives $0 \le T_1 \le \frac{1}{2}B_0 e\,(0.00652/B_0)^2 < 5.78 \times 10^{-5}/B_0$.

T_2 is the periodic-part difference: $|T_2| \le \sup|H'| \cdot |w^* - w_0| < 7.82 \times 10^{-6}/B_0$ by (5).

T_3 is the saddle-location shift itself, $|T_3| \le 0.00652/B_0$, again from Lemma 16.

T_4 compares the two normalizations. By $V(r^*) = g''_\tau(w^*)$ and the expression for g''_τ used in Lemma 16,

$$V(r^*) - \Bigl(B_0 - \frac{1}{c}\Bigr) = B_0\bigl(e^{w^*-w_0} - 1\bigr) + H''(w^*) + \Delta''(w^*),$$

whose modulus is at most $0.00652e^{0.00652/5} + \sup|H''| + 10^{-60} < 0.0135$, since $B_0|e^{w^*-w_0} - 1| \le B_0|w^* - w_0|e^{|w^*-w_0|}$. Dividing by $B_0 - 1/c$ and taking $\frac{1}{2}\log(1+\cdot)$ gives $|T_4| \le 0.0068/(B_0 - 1/c)$.

T_5 is the remainder of Theorem 4, evaluated at the true saddle rather than dropped. Here $B_0 \ge 5$ forces $w_0 \ge 5.3492$ and $w^* \ge w_0 - 0.0014$, so $e^{w^*} \ge 210.1$ and $|T_5| \le \sum_{k\ge0} 2e^{-2\cdot3^k\cdot210.1} < 10^{-182}$.

Each bound is $O(1/B_0)$ (for T_4, via $B_0 - 1/c \ge B_0 - 1$, comparable to B_0 once $B_0 \ge 5$). Finally $\tau = w_0 - \log B_0$ and $w_0 = c(B_0+a)$ give $\tau = cB_0 + ca - \log B_0$, so $B_0/\tau \to 1/c$, and $O(1/B_0) = O(1/\tau)$ with constants free of the phase of τ. ☐

6. The Berg–Krüppel identity

Proposition 18. *Write $f(p) = \alpha \log p - \beta \log^2 p$, so that Berg and Krüppel's transform $G_0(p) = p^\alpha \exp(-\beta \log^2 p)$ of their equation (9.3) is $e^{f(p)}$ and their equation (9.5) reads*

$$g_0(t) = \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} \exp[pt + f(p)]\, dp, \qquad \sigma > 0.$$

Take α, β from their equation (9.4) at $a = 3$, $\lambda = 2/3$. Then $f(p) = Q(\log p)$ identically; the exact saddle of that integral, $t + f'(x) = 0$ with $t = e^{-\tau}$, is $x^\sharp = e^{w_0(\tau)}$; and

$$P(\tau) \;=\; x^\sharp t + f(x^\sharp) - \tfrac{1}{2}\log\big(2\pi f''(x^\sharp)\big),$$

the saddlepoint expression invoked in the proof of their Proposition 9.1, evaluated at that exact saddle. In the variable $Y := cB_0$, with $r = -ca - \log c$, the saddle equation is $Y - \log Y = \tau + r$.

Proof. Their truncated equation (9.2) is $\lambda\psi'(t) = a\,\psi(at)$ for $a > 1$, $\lambda > 0$; Laplace transforming (the boundary term $\lambda\psi(0)$ from $\mathcal{L}[\psi'](p) = pG(p) - \psi(0)$ vanishes, since the solutions of interest are supported on $[0, \infty)$ with $\psi(0) = 0$) gives $\lambda p\, G(p) = G(p/a)$, whose general solution (9.3) is G_0 times an arbitrary 1-periodic function of $\log p / \log a$, with

$$\alpha = -\frac{1}{2} - \frac{\log \lambda}{\log a}, \qquad \beta = \frac{1}{2\log a}$$

their equation (9.4). Section 2 placed φ in that family at $a = 3$, $\lambda = 2/3$: their untruncated (9.1) reads $\frac{2}{3}\varphi'(t) = 3\big(\varphi(3t) - \varphi(3t-2)\big)$, which on $(0, \frac{2}{3})$ is $\varphi'(t) = \frac{9}{2}\varphi(3t)$. At those values

$$\alpha = -\frac{1}{2} - \frac{\log(2/3)}{\log 3} = \frac{1}{2} - \frac{\log 2}{\log 3} = a, \qquad \beta = \frac{1}{2\log 3} = \frac{1}{2c},$$

where the a on the right is this paper's linear coefficient of Q; their a, the dilation factor 3, is a different quantity and is not written again below. Hence, writing $y := \log p$,

$$f(p) = \alpha y - \beta y^2 = ay - \frac{y^2}{2c} = Q(y),$$

the first of the proposition's four claims. This is an identity in y, not merely a value match: the chain rule ($p\, d/dp = d/dy$) applied to f's first and second derivatives confirms it also at the level of Q' and Q'',

$$-p\, f'(p) = -Q'(y) = B_{\rm sm}(y) = 2\beta y - \alpha, \qquad p^2 f''(p) = Q''(y) - Q'(y) = B_{\rm sm}(y) - \frac{1}{c} = 2\beta y - 2\beta - \alpha.$$

Berg and Krüppel's own displayed formula for $f''(p)$, on p. 179 of [2], in the proof of their Proposition 9.1 (the unnumbered display right after "In view of"), reads $p^2 f''(p) = 2\beta \ln p + 2\beta - \alpha$, with $+2\beta$ where direct differentiation of f gives -2β: $f''(p) = \frac{1}{p^2}\big(2\beta \ln p - 2\beta - \alpha\big)$, since $f'(p) = \frac{1}{p}(\alpha - 2\beta \ln p)$ differentiates to $-\frac{2\beta}{p^2} - \frac{\alpha - 2\beta\ln p}{p^2}$. Their Proposition 9.1 is unaffected, the term being of lower order there than $2\beta \log p$, but the exact expression used here is not: the $\pm 2\beta$ discrepancy shifts $(x^\sharp)^2 f''(x^\sharp) = B_0 - 1/c$ by $2/c$, hence shifts $-\frac{1}{2}\log\big(2\pi(B_0 - 1/c)\big)$ in $P(\tau)$ below by $O(1/B_0)$, which vanishes as $B_0 \to \infty$.

The saddle condition $t + f'(x) = 0$ at $x = e^y$ now reads $e^{-\tau} = B_{\rm sm}(y)e^{-y}$, that is $\tau = y - \log B_{\rm sm}(y) = \Phi_0(y)$, whose unique solution on the branch $B_{\rm sm} > 1/c$ is $y = w_0(\tau)$ by Lemma 16. So $x^\sharp = e^{w_0}$, and there $x^\sharp t = e^{w_0 - \tau} = B_0$, $f(x^\sharp) = Q(w_0)$, $(x^\sharp)^2 f''(x^\sharp) = B_0 - 1/c > 0$. Substituting,

$$x^\sharp t + f(x^\sharp) - \tfrac{1}{2}\log\big(2\pi f''(x^\sharp)\big) = B_0 + Q(w_0) - \tfrac{1}{2}\log\frac{2\pi(B_0 - 1/c)}{(x^\sharp)^2}$$

$$= Q(w_0) + B_0 + w_0 - \tfrac{1}{2}\log\Big(2\pi\big(B_0 - \tfrac{1}{c}\big)\Big),$$

which is $P(\tau)$. Finally $w_0 = c(B_0 + a)$ and $\tau = w_0 - \log B_0$ give $\tau = cB_0 + ca - \log B_0 = Y - \log Y + ca + \log c = Y - \log Y - r$. $\square$

Series-reverting $Y - \log Y = \tau + r$ to second order: writing $W := \tau + r$, a first substitution $Y = W + \log W + \delta$ into $Y - \log Y = W$ gives, since $\log(1+u) = u + O(u^2)$ with $u = (\log W + \delta)/W$, $\delta = (\log W + \delta)/W + O(\log^2 W/W^2)$, so $\delta = \log(W)/W + O(\log^2 W/W^2)$ (the single term $Y = W + \log W$ alone is only accurate to $O(\log W/W)$, one order too coarse; this second term is needed). Hence $Y = W + \log W + \log(W)/W + O(\log^2 W/W^2)$, and $\log W = \log\tau + r/\tau + O(1/\tau^2)$ then gives $Y = \tau + \log\tau + r + (\log\tau + r)/\tau + O(\tau^{-2}\log^2\tau)$. Since $w_0 = Y + ca$ and $Q(w_0) = -Y^2/(2c) + ca^2/2$ (substituting $w_0 = Y + ca$ into Q's definition and simplifying), P's leading term in Y is $-Y^2/(2c)$, so $\partial P/\partial Y = -Y/c + O(1)$ turns the $O(\tau^{-2}\log^2\tau)$ error in Y above into the $O(\tau^{-1}\log^2\tau)$ error in P below. Substituting into P gives

$$P(\tau) = -\frac{(\tau + \log\tau)^2}{2c} + \Big(1 + \tfrac{1}{c} - \tfrac{r}{c}\Big)\tau + \Big(\tfrac{1}{2} - \tfrac{r}{c}\Big)\log\tau + C_P + O\bigg(\frac{\log^2\tau}{\tau}\bigg),$$

$$C_P = -\frac{r^2}{2c} + r + ca + \frac{ca^2}{2} - \frac{\log 2\pi}{2} + \frac{\log c}{2} = -0.95767431831339827606691224978558 55\ldots.$$

Berg and Krüppel's own equations following (9.6) give, in their notation,

$$\delta_{\mathrm{BK}} = \tfrac{1}{2} + \alpha - 2\beta\log(2\beta), \qquad \gamma = -2\beta - \delta_{\mathrm{BK}} - \tfrac{1}{2}, \qquad \varepsilon = \tfrac{1}{2} + \alpha - \beta\log(2\beta),$$

with $\alpha = a$, $\beta = 1/(2c)$ as identified above. Substituting and simplifying (using $r = -ca - \log c$) gives, matching the τ, $\log\tau$, and constant coefficients of P term for term,

$$\gamma = -\Big(1 + \tfrac{1}{c} - \tfrac{r}{c}\Big) = -1.8649154949\ldots, \qquad \delta_{\mathrm{BK}} = \tfrac{1}{2} - \tfrac{r}{c} = 0.4546762683\ldots,$$

$$\varepsilon = a + \tfrac{1}{2} + \frac{\log c}{2c} = 0.4118732574\ldots,$$

so that, in particular, $C_P = \varepsilon\log(2\beta) - \frac{1}{2}\log(2\pi)$ holds identically (verified by substituting ε's expression directly into C_P's), giving

$$C_P = \varepsilon\log(2\beta) - \tfrac{1}{2}\log(2\pi) = \log\frac{(2\beta)^\varepsilon}{\sqrt{2\pi}}.$$

Hence $P(\tau) - \log\varphi_{0,\mathrm{bare}}(e^{-\tau}) \to C_P$ and, using the prefactor-included normalization, $P(\tau) - \log\varphi_0(e^{-\tau}) \to 0$ with rate $O(\tau^{-1}\log^2\tau)$: the τ, $\log\tau$, and constant coefficients of P reproduce γ, δ_{BK}, ε exactly, so this convergence is a direct consequence of the identity of Proposition 18 together with the series reversion above, not an appeal to Berg and Krüppel's own asymptotic analysis of φ_0.

Proof of Corollary 3. By Proposition 17 and 18, $S(\tau) = \log\varphi_0(e^{-\tau}) + H(w_0(\tau)) + O(\tau^{-1}\log^2\tau)$, i.e. $\varphi_{\mathrm{sp}} = \varphi_0 \cdot e^{H(w_0(\tau))}(1+o(1))$; by Theorem 13, $\varphi = \varphi_{\mathrm{sp}}(1+o(1))$ as well, so $\varphi = \varphi_0 \cdot e^{H(w_0(\tau))}(1+o(1))$: the periodic factor multiplying Berg and Krüppel's asymptotic solution, in equation (9.7) for this eigenfunction, is e^H, matching the product formula their Proposition 9.3 supplies for it, as noted after Theorem 4. Proposition 6's Fourier series for H is the new, closed form; Proposition 8's certificate is what neither their product formula nor their closing remark (that they only *expect* φ bounded and bounded away from zero relative to φ_0) settles. $\square$

7. Proofs of the main theorems

Throughout this section, Theorem 13's parameter N and Sections 5–6's parameter τ are combined freely into one $o(1)$: since $s = e^{w^*}$ and, by Lemma 16, $w^* = w_0(\tau) + O(1/B_0)$ with $w_0(\tau) = c(B_0 + a)$, $N = \lfloor\log_3(2s)\rfloor \sim w^*/c \sim \tau/c$ as $\tau \to \infty$, so $N \to \infty$ exactly when $\tau \to \infty$. On $\widetilde{A}_\delta$ below, $\tau_l \to \infty$ uniformly, at a rate depending only on l and δ, not on the particular sequence chosen.

Proof of Theorem 1. Wirsching's class $\widetilde{A}_\delta$ (his equations (1.5) and (3.2)) consists of sequences (x_l) with $\lfloor 3^l x_l \rfloor = k_l$ satisfying $|l - k_l| \le \delta\sqrt{l}$ for every l; since $|3^l x_l - k_l| < 1$ as well, $|3^l x_l - l| < 1 + \delta\sqrt{l}$, hence $\lambda_l := 3^l x_l / l \to 1$ uniformly at rate $O(1/l) + O(\delta/\sqrt{l})$ on the class. Writing $z_l := \lambda_l\, l\, 3^{-l} = x_l$ (Wirsching's own z_l from the conjecture statement and his x_l from the class definition coincide, by definition of λ_l), $\tau_l = -\log z_l = lc - \log l - \log \lambda_l$, the saddle location satisfies $w_0(\tau_l) = lc - \log \lambda_l + O(1/l)$: Section 6's reversion of $Y - \log Y = \tau + r$ gives $w_0(\tau) = \tau + \log(\tau/c) + (\log \tau + r)/\tau + O(\tau^{-2}\log^2 \tau)$, and the naive one-term truncation $w_0(\tau) \approx \tau + \log(\tau/c)$ alone is only accurate to $O(\log \tau/\tau)$, one order too coarse; substituting $\tau = \tau_l$ into the full two-term formula, the $O(\log l/l)$ pieces from expanding $\log(\tau_l/c)$ against $\tau_l = lc - \log l - \log \lambda_l$ cancel exactly against the reversion's own $(\log \tau_l + r)/\tau_l$ term, leaving the stated $O(1/l)$, the same cancellation mechanism as the series reversion above. So $\operatorname{dist}(w_0(\tau_l), c\mathbb{Z}) \to 0$ uniformly on the class, at rate $O(1/\sqrt{l})$: writing $w_0(\tau_l) = lc + \varepsilon_l$ with $\varepsilon_l = -\log \lambda_l + O(1/l) \to 0$, the nearest multiple of c is lc itself once $|\varepsilon_l| < c/2$, so $H(w_0(\tau_l)) \to H(0)$ by continuity, regardless of the sign of ε_l. By Theorem 13, Proposition 17, Proposition 18 (which gives $P(\tau) - \log \varphi_0(e^{-\tau}) \to 0$, Section 6), and continuity of H (indeed Lipschitz, by (5)),

$$\log \varphi(z_l) - \log \varphi_0(z_l) = H(w_0(\tau_l)) + o(1) \to H(0)$$

uniformly on $\widetilde{A}_\delta$, i.e. $\varphi(z_l)/\varphi_0(z_l) \to e^{H(0)}$ uniformly. This is exactly Conjecture 3, with the constant given explicitly. □

Remark 19. The proof above uses only $\lambda_l \to 1$, equivalently $|l - k_l| = o(l)$; Wirsching's specific $O(\sqrt{l})$ bound in $\widetilde{A}_\delta$ plays no role beyond forcing this weaker condition. The same conclusion, and the same limit $e^{H(0)}$, holds for the broader class of sequences satisfying only $|l - k_l| = o(l)$.

Proof of Theorem 2. As $t = e^{-\tau}$ ranges over $(0, 1/2)$ with no restriction, $\tau \to \infty$. By Lemma 16, $w_0 = \Phi_0^{-1}$ is continuous and strictly increasing and unbounded on (τ_0, ∞), so $w_0(\tau) \bmod c$ sweeps every value in $[0, c)$ infinitely often as $\tau \to \infty$ (a continuous, strictly increasing, unbounded function hits every residue class modulo c infinitely often), so by the same combination of results used above, $\log \varphi(t) - \log \varphi_0(t) = H(w_0(\tau)) + o(1)$ takes values arbitrarily close to both $\sup H$ and $\inf H$ infinitely often. By Proposition 8, $\operatorname{osc}(H) > 0$ rigorously, so $\varphi(t)/\varphi_0(t)$ does not converge, and $\limsup / \liminf \ge e^{\operatorname{osc}(H)} \ge 1.0004188$: the ratio swings by at least 0.0418%. □

Proposition 20 (Wirsching's actual requirement)**.** *For $(x_l) \in \widetilde{A}_\delta$, write $x_l^+ := x_l + 3^{-(l+1)}$ (his equation (7.14); since $x_l \sim l \cdot 3^{-l}$, x_l^+ is asymptotic to x_l and plays the same role as this paper's z_l). Wirsching's own use of Conjecture 3, via $\varphi'(x) = \frac{9}{2}\varphi(3x)$ on $(0, \frac{2}{3})$ and his own calculation, his equation (7.13), that* $\lim_l (2/3^{l+1})\varphi_0'(x_l)/\varphi_0(x_l) = 2/3$ *uniformly on the class, reduces to*

$$\limsup_l \Lambda_l < \tfrac{3}{2}, \qquad \Lambda_l := \frac{\varphi(3x_l^+)/\varphi_0(3x_l^+)}{\varphi(x_l^+)/\varphi_0(x_l^+)}.$$

This holds with a wide margin: $\Lambda_l \to 1$.

Proof. First, the reduction itself. Wirsching's actual requirement is his inequality (7.5),

$$\limsup_l \frac{2}{3^{l+1}} \cdot \frac{\varphi'(x_l^+)}{\varphi(x_l^+)} < 1;$$

by $\varphi' = \frac{9}{2}\varphi(3\cdot)$ this is $\limsup_l (9/3^{l+1})\varphi(3x_l^+)/\varphi(x_l^+) < 1$. By definition of Λ_l, $\varphi(3x_l^+)/\varphi(x_l^+) = \Lambda_l \cdot \varphi_0(3x_l^+)/\varphi_0(x_l^+)$, and Berg and Krüppel's φ_0 is an asymptotic solution of the same truncated equation (their equation (7.12) in Wirsching's notation: $\varphi_0'(t)/\varphi_0(3t) \to 9/2$ as $t \to 0$), so $\varphi_0(3x_l^+)/\varphi_0(x_l^+) = \frac{2}{9}\varphi_0'(x_l^+)/\varphi_0(x_l^+)\,(1 + o(1))$. To see that (7.13) transfers from x_l to x_l^+ with an $o(1)$ correction, write $v = \log t$; the explicit form of φ_0 above gives $\varphi_0'(t)/\varphi_0(t) = B(v)/t$ with $B(v) = \gamma + \delta_{\mathrm{BK}}/v - 2\beta(v - \log(-v))(1 - 1/v)$, and, as $v \to -\infty$, $B(v) \sim -2\beta v$ while $B'(v) \to -2\beta$, so $(\log B)'(v) = B'(v)/B(v) \to 0$ and $(\log[\varphi_0'/\varphi_0])'(v) = (\log B)'(v) - 1 \to -1$, bounded. Since

$x_l^+/x_l = 1 + 3^{-(l+1)}/x_l \to 1$ (as $x_l \sim l \cdot 3^{-l}$, in fact $3^{-(l+1)}/x_l = O(1/l)$), $v(x_l^+) - v(x_l) = \log(x_l^+/x_l) = O(1/l)$, and the mean value theorem gives $\log[\varphi_0'/\varphi_0(x_l^+)] - \log[\varphi_0'/\varphi_0(x_l)] = O(1/l)$, i.e. $\varphi_0'(x_l^+)/\varphi_0(x_l^+) = \varphi_0'(x_l)/\varphi_0(x_l) \cdot (1 + O(1/l))$. Substituting and using his equation (7.13), $\lim_l (2/3^{l+1})\varphi_0'(x_l)/\varphi_0(x_l) = 2/3$ uniformly on the class,

$$\frac{9}{3^{l+1}} \cdot \frac{\varphi(3x_l^+)}{\varphi(x_l^+)} = \Lambda_l \cdot \frac{2}{3^{l+1}} \cdot \frac{\varphi_0'(x_l^+)}{\varphi_0(x_l^+)}\,(1 + o(1)) = \frac{2}{3}\,\Lambda_l \cdot (1 + o(1)),$$

so the requirement $\limsup_l (9/3^{l+1})\varphi(3x_l^+)/\varphi(x_l^+) < 1$ is exactly $\limsup_l \Lambda_l < 3/2$.

For the value of Λ_l: write $\tau_l := -\log x_l^+$ (locally to this proof only, distinct from the τ_l of Theorem 1's proof), so $t = x_l^+$ corresponds to $\tau = \tau_l$ and $t = 3x_l^+$ to $\tau = \tau_l - c$. The phase shift under $t \mapsto 3t$, i.e. $\tau \mapsto \tau - c$, is $w_0(\tau - c) - w_0(\tau) + c = O(1/\tau)$ by Lemma 16's defining equation, independently of the phase of τ; since H is c-periodic, $H(w_0(\tau - c)) = H(w_0(\tau - c) + c) = H(w_0(\tau) + O(1/\tau))$, and H Lipschitz (by (5)) turns this into $H(w_0(\tau - c)) = H(w_0(\tau)) + O(1/\tau)$. By Theorem 13, Proposition 17, and Proposition 18 together (the same chain used in the proof of Theorem 1), $\log\varphi(t) - \log\varphi_0(t) = H(w_0(\tau)) + o(1)$ uniformly in phase, so

$$\log \Lambda_l = H(w_0(\tau_l - c)) - H(w_0(\tau_l)) + o(1) = O(1/\tau_l) + o(1) \to 0,$$

using that H is Lipschitz ((5)) and the phase shift above is $O(1/\tau_l)$; hence $\Lambda_l \to 1$, well inside $\limsup \Lambda_l < 3/2$. □

8. Discussion

These results settle Conjecture 3 alone. Wirsching's positive-predecessor-density argument reduces to three conjectures; Conjectures 1 and 2, concerning respectively a pointwise-versus-average transfer for the Elka functions and a uniformity-near-a-singularity statement for the limiting operator S_∞, are untouched here, and the full theorem is not established by this paper.

Wirsching stated the weaker Conjecture 3 instead of the stronger $\varphi \sim \kappa\varphi_0$ he considers first. Theorem 2 shows why: the stronger statement fails by a mechanism with a certified, positive amplitude. Every phase $\theta \in [0, c)$ swept by H admits some sequence along which the ratio converges to $e^{H(\theta)}$: take $\tau_l := \Phi_0(\theta + lc)$, $z_l := e^{-\tau_l}$, so $w_0(\tau_l) = \theta + lc$ exactly, and apply Theorem 13, Proposition 17, and Proposition 18 as in the proof of Theorem 1. What is special about phase 0 is only that it is the one Wirsching's own class $\widetilde{A}_\delta$ happens to fix. That phase sits close to $w_{1011118}$, one of four points where H is printed in Proposition 8's proof: combining that value with $H(0)$ gives $H(0) - H(w_{1011118}) = 5.25 \times 10^{-6}$, a floating-point computation (not itself a certified enclosure), against a certified oscillation of at least 4.18×10^{-4}. (Whether $w_{1011118}$ is where H attains its true minimum is not certified, per Proposition 8's own caveat on the grid search.) We do not know whether the near-match with phase 0 is coincidental.

The rate is not established beyond $o(1)$. The sharper Edgeworth correction of Remark 15, and a correspondingly sharper form of the envelope and Berg–Krüppel comparisons, would give an explicit polynomial rate throughout; neither theorem needs it.

9. Code and data availability

Every certified or numerically-checked claim in this paper is backed by a script in the accompanying repository, https://github.com/faculdade/wirsching-conjecture3-proof, organized by section with a README per script stating what it verifies and how to run it. This includes the interval-arithmetic certificates behind Proposition 8 and the bounds (5), the constant chain behind Theorem 13, and the identity of Proposition 18.

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